1 F^n

Recall that F stands for $\mathbb{R}$ or $\mathbb{C}$.

Definition 1.1 (F^n)**.** For each positive integer n, define F^n to be the set of all ordered n-tuples of elements of F. That is,

$$F^n := \{(x_1,\dots,x_n) : x_1,\dots,x_n \in F\}.$$

We often denote an element of F^n by a single letter, such as x. That is, we let

$$x = (x_1,\dots,x_n).$$

Then $x_k \in F$ is called the *k-th component* (or kth *coordinate)* of x. Two elements of F^n are *equal* if and only if the corresponding coordinates are all equal. That is, $(x_1,\dots,x_n) = (y_1,\dots,y_n)$ if and only if $x_1 = y_1,\dots,x_n = y_n$.

Example 1.2.

(a) An element of F^1 is a 1-tuple (x), with $x \in F$. We identify (x) with x, so that $F^1 = F$.

(b) $\mathbb{R}^2 = \{(x,y) : x,y \in \mathbb{R}\}$ is the plane.

(c) $\mathbb{R}^3 = \{(x,y,z) : x,y,z \in \mathbb{R}\}$ is three-dimensional space.

(d) $\mathbb{C}^4 = \{(z_1,z_2,z_3,z_4) \in \mathbb{C}\}$ is the set of all ordered 4-tuples of complex numbers.

Definition 1.3 (Addition in F^n)**.** *Addition* in F^n is defined by adding the corresponding components:

$$(x_1,\dots,n) + (y_1,\dots,y_n) = (x_1+y_1,\dots,x_n+y_n).$$

Proposition 1.4. *Addition in F^n has satisfies the same properties as addition in F, namely*

Properties of Addition
(A1) *(Commutativity) $x+y = y+x$ for all $x,y \in F^n$.*
(A2) *(Associativity) $(x+y)+z = x+(y+z)$ for all $x,y,z \in F^n$*
(A3) *(Identity) There exists an element $0 \in F^n$ such that $0+x = x$ for every $x \in F^n$.*
(A4) *(Additive Inverses) To each $x \in F^n$ there exists an element $-x \in F^n$ such that $x+(-x) = 0$.*

Proof. To prove (A1), let $x = (x_1,\dots,x_n)$ and $y = (y_1,\dots,y_n)$. Then

$$\begin{aligned} x+y &= (x_1+y_1,\dots,x_n+y_n) \\ &= (y_1+x_1,\dots,y_n+x_n) \qquad \text{(since (A1) holds for each } x_k+y_k \in F) \\ &= y+x. \end{aligned}$$

The proof that the remaining properties hold is left as an exercise. □

1. Complete the proof of Proposition 1.4.
   (A2):

$$(x+y)+z=((x_1+y_1)+z_1,\ldots,(x_n+y_n)+z_n)=(x_1+(y_1+z_1),\ldots,x_n+(y_n+z_n))=x+(y+z).$$

   (A3): $0=(0,\ldots,0)$
   (A4): $-x=(-x_1,\ldots,-x_n)$.

Note that the proofs of the properties which follow from (A1)-(A4), such uniqueness of the additive identity 0, the cancellation law, etc., all carry over immediately to F^n.

Let's now turn to multiplication. Suppose we try to define multiplication on F^n similar to the way we defined addition, namely by multiplying the corresponding coordinates together. That is,

$$(x_1,\ldots,x_n)(y_1,\ldots,y_n)=(x_1y_1,\ldots,x_ny_n).$$

This is clearly commutative and associative, and the multiplicative identity is $(1,\ldots,1)$. However, not all nonzero elements of F^n have a multiplicative inverse. For example, $(1,0,\ldots,0)\neq(0,\ldots,0)=0$, but there is no element $(x_1,\ldots,x_n)\in F^n$ such that $(x_1,\ldots,x_n)(1,0,\ldots,0)=(x_1,0,\ldots,0)=(1,1,\ldots,1)$. Thus, this definition of multiplication does not make F^n into a field. For this reason, and others, the definition above turns out to not be useful. What does turn out to be useful, is to define the product of an element of F^n by an element of F.

Definition 1.5 (Product of an element of F and an element of F^n)**.** The product of $\alpha\in F$ and $x=(x_1,\ldots,x_n)\in F^n$ is defined by multiplying each component of x by α. That is,

$$\alpha(x_1,\ldots,x_n):=(\alpha x_1,\ldots,\alpha x_n)$$

2. Prove that, for all $\alpha,\beta\in F$ and all $x,y\in F^n$, the following properties are satisfied:

   (a) $(\alpha\beta)x=\alpha(\beta x)$
   Write $x=(x_1,\ldots,x_n)$. Then

$$\begin{aligned}(\alpha\beta)x=(\alpha\beta)(x_1,\ldots,x_n)&=((\alpha\beta)x_1,\ldots,(\alpha\beta)x_n)\\&=(\alpha(\beta x_1),\ldots,\alpha(\beta x_n))\qquad\text{(by (M2) in } F)\\&=\alpha(\beta x).\end{aligned}$$

   (b) $(\alpha+\beta)x=\alpha x+\beta x$

$$\begin{aligned}(\alpha+\beta)x&=(\alpha+\beta)(x_1,\ldots,x_n)\\&=((\alpha+\beta)x_1,\ldots,(\alpha+\beta)x_n)\\&=(\alpha x_1+\beta x_1,\ldots,\alpha x_n+\beta x_n)\qquad\text{(by (D) in } F)\\&=\alpha x+\beta x\end{aligned}$$

(c) $\alpha(x+y) = \alpha x + \alpha y$

$$\begin{aligned}\alpha(x+y) &= \alpha(x_1+y_1,\ldots,x_n+y_n)\\ &= (\alpha(x_1+y_1),\ldots,\alpha(x_n+y_n))\\ &= (\alpha x_1+\alpha y_1,\ldots,\alpha x_n+\alpha y_n) \qquad \text{(by (D) in } F)\\ &= \alpha x+\alpha y\end{aligned}$$

(d) $1x = x$

$$\begin{aligned}1x = 1(x_1,\ldots,x_n) &= (1x_1,\ldots,1x_n)\\ &= (x_1,\ldots,x_n) \qquad \text{(by (M3) in } F)\\ &= x.\end{aligned}$$

3. Find $x \in \mathbb{R}^4$ such that $(4,-3,1,7)+2x = (5,9,-6,8)$.
Subtract $(4,-3,1,7)$ from both sides of the equation:

$$\begin{aligned}2x &= (5,9,-6,8)-(4,-3,1,7)\\ 2x &= (5-4,9+3,-6-1,8-7)\\ 2x &= (1,12,-7,1).\end{aligned}$$

Now multiply both sides by $\frac{1}{2}$:

$$\begin{aligned}x &= \frac{1}{2}(1,12,-7,1)\\ &= \boxed{\left(\frac{1}{2},6,-\frac{7}{2},\frac{1}{2}\right)}.\end{aligned}$$

4. Explain why there does not exist $\alpha \in \mathbb{C}$ such that

$$\alpha(2-3i,5+4i,-6+7i) = (12-5i,7+22i,-32-9i).$$

The equation

$$(\alpha(2-3i),\alpha(5+4i),\alpha(-6+7i)) = (12-5i,7+22i,-32-9i)$$

implies

$$\begin{aligned}\alpha(2-3i) &= 12-5i\\ \alpha(5+4i) &= 7+22i\\ \alpha(-6+7i) &= -32-9i.\end{aligned}$$

The first equation implies

$$\alpha = \frac{(12-5i)}{(2-3i)} \cdot \frac{(2+3i)}{(2+3i)} = \frac{24+15+(-10+36)i}{13} = \frac{39+26i}{13} = 3+2i.$$

With this value of α,

$$(3+2i)(5+4i) = 15-8+(10+12)i = 7+22i,$$

so the second equation holds. However,

$$(3+2i)(-6+7i) = -18-14+(-12+21)i = -32+9i \neq -32-9i,$$

so the third equation is not satisfied. Hence, there is no such α.